

Speakers

C. Muthukrishnan,
CEO, Semiconductor
Fabless Accelerator
Lab (SFAL)



Hemant Mallapur,
co-founder and
executive VP,
Engineering,
Saankhya Labs

Dr PK. Sundararajan,
founder and CEO,
BluArmor



Poomima
Shenoy, CEO,
Hummingbird

Sanjeev Kesar,
managing director,
Arrow Electronics
India



Tarun Verma,
managing partner,
Silicon Catalyst

The right architecture for becoming techno-commercially successful, as per Verma, is a must. A wonderful product, as per him, is the one that creates value both for the customers as well as for the creators. Everything else, he thinks, is either a white elephant or a product not fit commercially at all.

This is exactly where startups make or break the line. Verma elaborates that reading the market for a startup should be of utmost importance. Teams, as per him, should pos-

Where Do the Big Opportunities Lie?

- **AI, ML, and Deep Learning**—Not many low-level infrastructure IPs have been created. These verticals have a lot of scope for startups
- **5G is just starting**—The boom for semiconductors required for the 5G industry will be big. This is one more area where startups can focus on
- **Telehealth and contactless products** are going to stay here for a long time, so one more area for startups to focus on
- **The energy-saving segment**, as the world is becoming environmentally cautious
- **Another segment is ADAS (Assisted Driving)**, as the trend is just picking up. Tesla has already changed the automotive game in the USA
- **Automation is going to be adopted at a very fast pace**. Designing solutions in this vertical should also be a focus of startups
- **IoT**—As everything becomes connected, the world will require a lot of connectivity modules. Even the smart meters need to be connected via IoT. Industrial, home, or government, almost every device will be connected with IoT modules. Designing these can be a focus area for startups

sess the analytical ability to assess the market and present a detailed report to investors as part of the pitch they are making. Verma also recommends formation of teams as a critical step. Like Kesar, he thinks there should be more than a single founder in a startup. "More co-founders help investors assess risk in a better manner," notes Verma.

Checklist before venturing into the space

Once not so popular vertical of electronics design and manufacturing in India has been made popular by initiatives like 'Make in India'. The recent call for 'Aatmanirbhar Bharat' has also made the vertical more enticing. The recent announcement of PLI schemes worth 500 billion rupees is a clear indicator of opportunities that will be created by big players like the Samsungs and Foxconn for MSMEs and startups in the near future.

"While a lot of big names are supporting these initiatives, it is critical to note that these big names support startups that have India as well as their global plans ready to be implemented," opines Shenoy.

The space also includes a lot of competition from within and outside India. In fact, the competition may be in its nascent stage inside the country; outside it is humongous. Countries like China are known for facilitating electronics manufacturing and design services at competitive prices. Similarly, when it comes to high-end electronics like semiconduc-

tor chips, countries like Taiwan lead the race.

"A great product can outdo any competition. The features of a great product include its ability to be developed into standards that are prevalent in different parts of the world," explains Verma.

Atomberg Technologies makes for a perfect example of a startup in the electronics design and manufacturing startup space in the country. The company recently raised 700 million rupees in funding during round B. Into designing and manufacturing BLDC-powered ceiling fans in India, the startup is now looking to venture into new consumer electronic verticals. Its strength lies in inhouse designing and manufacturing.

Saankhya Labs is another example of a successful startup when it comes to high-tech startups in the field of electronics design. It designs semiconductors for high-end applications. Though its end products are usually manufactured outside India, it has also found most of its clients outside the country.

"Success in the high-tech business is a mix of three things—the value of your idea; your ability to build it; and your ability to sell it." If not ten-on-ten, each of these must be able to score a decent value in the startup report card," says Hemant Mallapur, co-founder and executive VP, Engineering, Saankhya Labs. **EFY**

This article is based on inputs shared by industry leaders during panel discussions held at recent India Technology Week sessions